

Apply for a Science Corps Fellowship - Entry #3316

Name

Elizabeth Hayman

Email

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Gender

Female

City, state, and country of residence

Oxford, Oxfordshire, UK

Nationality

British

When did or will you finish your PhD?

October 2025

University Education

PhD | Generative Design of Hydrocephalus Shunts | University of Oxford | October 2025

Undergraduate Masters | Mathematics | University of Oxford | August 2021

What are your reasons for applying to be a Science Corps fellow?

During my time at Oxford, I've been involved in various teaching and outreach activities beyond my PhD, which I've found highly rewarding. The Science Corps fellowship stands out to me as a meaningful opportunity to continue sharing my love for STEM education while making a measurable impact.

A particularly valuable experience was being invited by the mathematics department to design and lead an outreach session for talented 14-16 year-

olds. It was an exciting challenge to design a fun and informative session that would be accessible to this age group while going beyond the school curriculum. I created an interactive activity where students used dice rolls to model the random movement of a smoke particle, then applied probability concepts to uncover expected patterns. The session was a success, and I was invited to repeat it the following year.

In addition to this, I've led undergraduate tutorials, planned outreach events for under-privileged sixth form students, and mentored students from underperforming backgrounds through their university applications. Having benefited from, and enjoyed, immensely my own academic journey, I am passionate about inspiring others and believe the Science Corps fellowship will allow me to further contribute to equitable education while continuing to grow as an educator.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

In addition to my PhD studies, I have held an Oxford college lectureship position for three years, specializing in courses for the first year undergraduate mathematicians. My role involves leading weekly small group classes to confirm and extend the students' knowledge of the syllabus, and ensure they are well prepared for exams. The small group size means I come to know the students intimately and adjust the teaching experience to their individual needs.

As part of this role I am involved in the interview process for incoming undergraduates giving me a detailed knowledge of the Oxbridge application process. I'm eager to share this expertise with students interested in applying to competitive universities, helping them navigate their own application journeys.

Outside of academia, I've held leadership roles in extracurricular activities, such as captaining the university mountain bike club since May 2023 and co-founding the university ice skating society, where I served as treasurer. Both roles involved organizing and leading training sessions, often introducing beginners to the sport and adjusting activities to accommodate varying skill levels. These experiences have enhanced my capacity to teach across a wide range of abilities, a skill I would bring to the Science Corps fellowship.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

Doing science in industry

In science education

In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

I am now starting my last year of my PhD, focused on finishing and writing up this project while continuing my teaching responsibilities. I intend to submit my thesis and complete the viva examination by October 2025. I would be immensely privileged to hold a Science Corps fellowship over winter 2025 and spring 2026. This experience would allow me to gain valuable insights into education across cultures while continuing to contribute to equitable access to STEM education. Additionally, I would strengthen my ability to adapt teaching techniques to diverse students, laying the groundwork for a long-term career that balances research with education.

I am investigating postdoctoral positions using my supervisors' networks to search for projects close to my skills and interests. In the next ten years, I aim to secure a permanent academic role that integrates research and teaching responsibilities, working to make mathematics and science more accessible to students from all backgrounds. I hope to play a key role in developing outreach initiatives and policies that promote inclusivity in science education while maintaining an active research profile. I aspire to influence how science is taught and communicated, making it more engaging and accessible to future generations.

CV/Resume[EHayman CV ScienceCorps.pdf](#)**How did you find out about Science Corps?**

University email list

Any other comments/questions?

Empty

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